

Software Requirements Specification

(SRS)

Project Name	HEALTH AI Co-Creation & Innovation Platform
Course	SENG 384 – Software Project IV
Group No / Name	WebStar
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Revision History

Date	Version	Change Description	Author
16/04/2026	1.0	Initial version	Sena Nur Pekgöz Ceren Kızılay
06/05/2026	2.0	Updated version	Sena Nur Pekgöz

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the HEALTH AI Co-Creation & Innovation Platform. This SRS serves as the primary guide for the development team and stakeholders to ensure all system objectives are met.

1.2 Scope

The platform enables structured partner discovery between healthcare professionals and engineers to foster multidisciplinary innovation. It provides a secure matchmaking infrastructure but does not provide financial transactions, contract management, medical advice, or file-sharing services.

1.3 Definitions and Abbreviations

Term / Abbreviation	Definition
SRS	Software Requirements Specification
SDD	Software Design Document
RBAC	Role-Based Access Control
GDPR	General Data Protection Regulation
NDA	Non-Disclosure Agreement
WCAG	Web Content Accessibility Guidelines
SPA	Single-Page Application (React frontend with client-side routing)
.edu	Institutional academic email domain used as a registration gate
Post	A structured collaboration announcement created by a registered user
Meeting Request	A formal request to initiate collaboration, requiring NDA acceptance
Lifecycle	The sequence of status changes a Post or Meeting Request passes through

1.4 Intended Audience

This document is intended for the WebStar development team, course instructors, and the Erasmus+ HEALTH-AI project evaluators.

2. Overall Description

2.1 Product Perspective

Multidisciplinary innovation in healthcare currently relies on coincidence or personal networks, which is inefficient. HEALTH AI addresses this gap by providing a structured, announcement-based discovery platform that eliminates randomness in interdisciplinary collaboration. The system is implemented as a React Single-Page Application (SPA) using TypeScript and localStorage for prototype-stage data persistence.

2.2 User Roles

Role	Description	Key Permissions
Engineer	Technical expert seeking clinical domain knowledge.	Register, create/edit own posts, browse posts, send meeting requests, view meeting requests on own posts.
Healthcare Professional	Medical expert with innovation ideas lacking technical infrastructure.	Register, create/edit own posts, browse posts, send meeting requests, accept/decline incoming requests.
Admin	System overseer responsible for platform integrity.	All user abilities plus: view all users, suspend/unsuspend accounts, remove any post, view full audit log.

2.3 Assumptions and Dependencies

- Users are assumed to have a valid institutional .edu email address.
- The system depends on an external email verification API (configured via VITE_AUTH_API_URL) for sending and confirming OTP codes.
- Prototype data is stored in browser localStorage; production deployment requires a persistent server-side database.
- External meeting tools (Zoom, Teams) handle actual video collaboration after the initial meeting request is accepted.

2.4 Constraints

- No Patient Data: The system shall not store patient data or confidential technical files.
- Text-Only Posts: Post creation is restricted to structured text fields; file uploads are not permitted.
- First Contact Only: The platform facilitates the initial match and meeting scheduling; it does not manage the collaboration itself.
- Prototype Scope: The current implementation is frontend-only with localStorage persistence; no production backend is deployed.

3. Functional Requirements

3.1 User Registration & Access Control

ID	Requirement Description	Priority	Source
FR-01	The system shall only allow registration with institutional .edu email addresses. A regex check is applied client-side (isInstitutionalEduEmail()) and validated before account creation.	High	Brief 4.1
FR-02	The system shall implement an email verification mechanism. After registration, users receive a one-time code via the external email API. Users must verify before logging in.	High	Brief 4.1
FR-03	The user shall be able to select a role during registration: Engineer or Healthcare Professional. The Admin role is pre-seeded and not self-assignable.	High	Brief 4.1
FR-04	An Admin shall be able to suspend or unsuspend any user account. Suspended accounts cannot log in and are blocked at session validation time.	High	Brief 4.1

3.2 Post Management

ID	Requirement Description	Priority	Source
FR-10	Authenticated users shall be able to create a structured post. Required fields: title, working domain, short explanation, desired expertise, commitment level, collaboration type, confidentiality level, project stage, country, city, and expiry date.	High	Brief 4.2
FR-11	Posts shall track the following lifecycle statuses: Draft, Active, Meeting Scheduled, Partner Found (Closed), Expired. Status transitions are recorded in a per-post lifecycle log.	High	Brief 4.2
FR-12	The system shall strictly prohibit file uploads. All post content is text-only. A warning is displayed when creating a post reminding users not to share confidential technical material.	High	Brief 4.2
FR-13	Posts shall support an automatic expiry date. When the expiry date is reached, the system automatically transitions the post status from Active to Expired. Auto-close behavior is configurable per post.	Medium	Brief 4.2
FR-14	A post owner shall be able to manually mark a post as "Partner Found" (Closed) at any time when the post is Active or Meeting Scheduled.	Medium	Brief 4.2

3.3 Search & Matching

ID	Requirement Description	Priority	Source
FR-20	Authenticated users shall be able to filter posts by: working domain (contains), city (exact), country (exact), project stage, post status, and free-text keyword (searches title, domain, explanation, expertise, city, country).	High	Brief 4.3
FR-21	The system shall visually highlight posts that match the user's entered city filter with a visual indicator (amber badge), providing city-based co-location awareness.	Medium	Brief 4.3

3.4 Meeting Request Workflow

ID	Requirement Description	Priority	Source
FR-30	An authenticated user shall be able to send a meeting request to the owner of any Active post. The request must include: NDA acceptance (mandatory checkbox) and at least one proposed time slot. A user cannot send a request to their own post.	High	Brief 4.4
FR-31	The post owner shall be able to accept, decline, or cancel a pending meeting request. On acceptance, the owner may select a preferred time slot from the proposed slots. Accepting a meeting request transitions the corresponding post status to "Meeting Scheduled".	High	Brief 4.4
FR-32	The requester shall be able to cancel a pending meeting request they sent. Cancellation is only allowed while the request is still in "pending" status.	Medium	Brief 4.4

3.5 Administrative Dashboard

ID	Requirement Description	Priority	Source
FR-40	Admin users shall have access to a dedicated dashboard displaying platform-wide statistics: total users, total posts, active posts, and total meeting requests.	High	Brief 4.5
FR-41	Admin users shall be able to view a complete list of registered users including name, email, role, verification status, and suspension status. Admins can suspend or unsuspend any non-admin account.	High	Brief 4.5
FR-42	Admin users shall be able to view the full activity audit log. The log shall be filterable and shall display: timestamp (at), user ID, role, action type, target entity, and result (success/failure). Admins shall be able to remove any post from the platform.	High	Brief 4.5

3.6 Activity Logging & Audit Trail

ID	Requirement Description	Priority	Source
FR-50	The system shall automatically log all security-sensitive actions. Logged action types include: login, logout, failed_login, register, verify_email, post_create, post_edit, post_status_change, meeting_request_create, meeting_request_update, admin_post_remove, admin_user_suspend, admin_user_unsuspend, security_event.	High	Brief 4.6
FR-51	Each audit log entry shall record: a unique ID, ISO timestamp (at), userId (optional for failed logins), role, actionType, targetEntity (affected post/meeting ID), result (success or failure), and optional details field for additional context.	High	Brief 4.6

4. Non-Functional Requirements

ID	Requirement	Metric / Target	Category
NFR-01	Search / filter results shall be returned within a specified time.	< 1.5 seconds (client-side filtering)	Performance
NFR-02	Page load time shall be limited for the initial application bundle.	< 3 seconds on standard broadband	Performance
NFR-03	The platform value proposition shall be understandable within a short time for a first-time visitor.	Purpose understood in ≤ 20 seconds	Usability
NFR-04	The system shall support GDPR rights: "Right to Delete" (account deletion) and "Right to Export" (data portability).	User can request deletion or export of personal data	Data Privacy
NFR-05	Audit logs shall be tamper-resistant and retained for a defined period.	Logs retained for 24 months (production); 2,000-entry cap in prototype	Audit / Security
NFR-06	Registration shall be restricted to institutional email addresses only.	100% of accounts must have a verified .edu address	Security
NFR-07	The UI shall be accessible and responsive across major browsers and screen sizes.	WCAG 2.1 AA compliance target; responsive from 320 px	Accessibility

5. Use Cases

5.1 UC-01: Engineer Creates a Post

Name	UC-01: Engineer Creates a Post
Actor(s)	Engineer (or Healthcare Professional)
Precondition	The user must be logged in and email-verified.
Main Flow	1. The user navigates to "Create announcement" from the navigation bar or the Announcements page. 2. The user fills in all required fields: title, working domain, short explanation, desired expertise, commitment level, high-level idea, collaboration type, confidentiality level, project stage, country, city, and expiry date. 3. The user selects the initial status (Draft or Active). 4. The user submits the form. The system validates all required fields. 5. The system saves the post via createPost() and records a post_create audit log entry. 6. The user is redirected to the Post Detail page.
Postcondition	The post is saved in localStorage and visible in the Announcements feed (if Active).
Alternative Flow	2a. If a required field is left blank, the system displays an inline error message and prevents submission.

5.2 UC-02: Requesting a Meeting

Name	UC-02: Requesting a Meeting
Actor(s)	Engineer or Healthcare Professional
Precondition	User is logged in. The target post must be in Active status. The user must not be the post owner.
Main Flow	1. User browses the Announcements page and selects an interesting post. 2. User clicks "Request Meeting" on the Post Detail page. 3. A meeting request panel opens. The user reads and accepts the NDA checkbox (mandatory). 4. The user proposes at least one time slot using the date/time input. 5. The user submits the request. The system validates NDA acceptance and at least one slot. 6. The system saves the meeting request via createMeetingRequest() and records a meeting_request_create audit log entry. 7. The request appears in the post owner's "Pending meeting requests" section.
Postcondition	A meeting request with "pending" status is created. The post owner is notified via the dashboard.
Alternative Flow	3a. If the NDA is not accepted, the Submit button remains disabled.4a. If no time slot is proposed, the system displays an error on submission.

5.3 UC-03: Post Owner Responds to a Meeting Request

Name	UC-03: Post Owner Responds to a Meeting Request
Actor(s)	Engineer or Healthcare Professional (as post owner)
Precondition	Post owner is logged in. At least one pending meeting request exists for one of their posts.
Main Flow	1. Post owner navigates to "My Announcements" or the Post Detail page. 2. The system displays pending meeting requests with requester name, time slots, and related post title. 3. The post owner clicks the meeting request to view the full proposal. 4. The post owner selects "Accept" and optionally picks a preferred time slot from the proposed options. 5. The system updates the meeting request status to "accepted" via <code>updateMeetingRequest()</code> and transitions the post status to "Meeting Scheduled". 6. Audit log entries are recorded for <code>meeting_request_update</code> and <code>post_status_change</code>.
Postcondition	The meeting request status is "accepted". The post status is "Meeting Scheduled". The agreed time slot is stored.
Alternative Flow	4a. If the post owner selects "Decline", the meeting request status is updated to "declined". The post remains Active. 4b. If the post owner selects "Cancel", the request status is updated to "cancelled".

5.4 UC-04: Admin Moderates the Platform

Name	UC-04: Admin Moderates the Platform
Actor(s)	Admin
Precondition	Admin is logged in.
Main Flow	1. Admin navigates to the Admin Panel via the navigation bar. 2. The Admin Panel displays platform statistics: total users, posts, active posts, meeting requests. 3. The admin reviews the User Management tab to see all registered users. 4a. To suspend a user: the admin clicks "Suspend" next to the user. The system sets <code>suspended=true</code> and records an <code>admin_user_suspend</code> audit event. The suspended user is immediately blocked from logging in. 4b. To remove a post: the admin finds the post in the Posts tab and clicks "Remove". The system calls <code>removePostAdmin()</code> and records an <code>admin_post_remove</code> audit event. 5. The admin reviews the Audit Logs tab to inspect security-sensitive activities.
Postcondition	The chosen action (suspension or post removal) is persisted. The audit log is updated.
Alternative Flow	4a. Unsuspend: the admin clicks "Unsuspend" to restore login access and records an <code>admin_user_unsuspend</code> event.

6. Data Model

The following entities are stored in the platform database (browser localStorage in the prototype; PostgreSQL in production). All entities are referenced by prefixed string IDs (e.g., usr_XXX, pst_XXX, mtg_XXX, log_XXX).

Entity	Key Fields	Relationships
User	id, email (.edu), name, role (engineer healthcare admin), verified (boolean), suspended (boolean), createdAt	Creates N Posts; sends N MeetingRequests; has N AuditLog entries.
Post	id, ownerUserId (FK → User), ownerRole, title, workingDomain, shortExplanation, desiredExpertise, commitmentLevel, highLevelIdea, collaborationType, confidentialityLevel, projectStage, country, city, status (draft active meeting_scheduled closed expired), expiryDate, autoClose, createdAt, updatedAt, lifecycle[]	Belongs to 1 User; has N MeetingRequests; lifecycle is an array of {at, byUserId, to} transitions.
MeetingRequest	id, postId (FK → Post), fromUserId (FK → User), toUserId (FK → User), ndaAccepted (boolean, must be true), proposedSlots (string[]), selectedSlot?, status (pending accepted declined cancelled), createdAt, updatedAt	Connects 2 Users via 1 Post.
AuditLog	id, at (ISO timestamp), userId? (FK → User), role?, actionType, targetEntity?, result (success failure), ipAddress?, details?	Tracks 1 User action; userId is optional (e.g., for failed_login).

7. Interface Requirements

7.1 User Interface (UI)

Screen	Description
LandingPage	Public home page. Communicates the platform's purpose within 20 seconds. Provides "Register" and "Log in" call-to-action buttons.
LoginPage	Email-only login form. Shows demo account hints for evaluators.
RegisterPage	Registration form with name, institutional .edu email, and role selection (Engineer / Healthcare Professional).
VerifyEmailPage	OTP verification screen. Calls the external email verification API to confirm the user's address before activating the account.
PostsPage (Dashboard)	Main announcement feed. Filterable by keyword, domain, city, country, project stage, and status. City-matched posts are highlighted with an amber badge. Includes "Create announcement" button.
PostEditorPage	Form for creating and editing structured posts. All fields are text-only. Includes confidentiality level and collaboration type selectors.

Screen	Description
PostDetailPage	Full view of a single post. Shows all public fields. Displays the "Request Meeting" panel (with NDA checkbox and time slot picker) for non-owners. Shows pending requests and action buttons for the post owner.
AdminPage	Admin-only dashboard. Tabs: Overview (statistics), Users (list with suspend/unsuspend), Posts (list with remove), Audit Logs (filterable log table).

7.2 External System Interfaces

- **Email Verification API:** A REST endpoint (configured via VITE_AUTH_API_URL) providing two routes: POST /auth/send-code (sends OTP to the user's email) and POST /auth/verify-code (validates the submitted OTP and activates the account).
- **External Meeting Tools:** After a meeting request is accepted, users are expected to coordinate via external tools (Zoom, Microsoft Teams). The platform generates no calendar events or links.

8. Requirements Traceability Matrix

Req. ID	Requirement Summary	Source (Brief)	Related Use Case(s)
FR-01	.edu email restriction on registration	Brief 4.1	UC-01, UC-02
FR-02	Email verification mechanism (OTP)	Brief 4.1	UC-01
FR-03	Role selection during registration	Brief 4.1	UC-01
FR-04	Admin can suspend / unsuspend accounts	Brief 4.1	UC-04
FR-10	Structured post creation (text-only fields)	Brief 4.2	UC-01
FR-11	Post lifecycle status tracking	Brief 4.2	UC-01, UC-03
FR-12	File upload prohibition	Brief 4.2	UC-01
FR-13	Automatic post expiry on expiry date	Brief 4.2	UC-01
FR-14	Manual "Partner Found" marking by post owner	Brief 4.2	UC-03
FR-20	Multi-criteria post filtering (domain, city, stage...)	Brief 4.3	UC-02
FR-21	City-based match highlighting	Brief 4.3	UC-02
FR-30	Meeting request with mandatory NDA acceptance	Brief 4.4	UC-02
FR-31	Post owner accept / decline / cancel requests + slot selection	Brief 4.4	UC-03
FR-32	Requester can cancel a pending request	Brief 4.4	UC-02
FR-40	Admin dashboard with platform statistics	Brief 4.5	UC-04
FR-41	Admin user management (view, suspend, unsuspend)	Brief 4.5	UC-04

Req. ID	Requirement Summary	Source (Brief)	Related Use Case(s)
FR-42	Admin audit log viewer + post removal	Brief 4.5	UC-04
FR-50	Automatic logging of all security-sensitive actions	Brief 4.6	UC-01 – UC-04
FR-51	Structured audit log entry schema (at, userId, role, result...)	Brief 4.6	UC-01 – UC-04
NFR-01	Search results < 1.5 s	Brief NFR	—
NFR-02	Page load < 3 s	Brief NFR	—
NFR-03	Purpose understandable in ≤ 20 s	Brief NFR	—
NFR-04	GDPR: Right to Delete & Export	Brief NFR	UC-04
NFR-05	Tamper-resistant logs retained 24 months	Brief NFR	UC-04
NFR-06	.edu-only registration enforced 100%	Brief NFR	UC-01
NFR-07	WCAG 2.1 AA / responsive 320 px+	Brief NFR	—

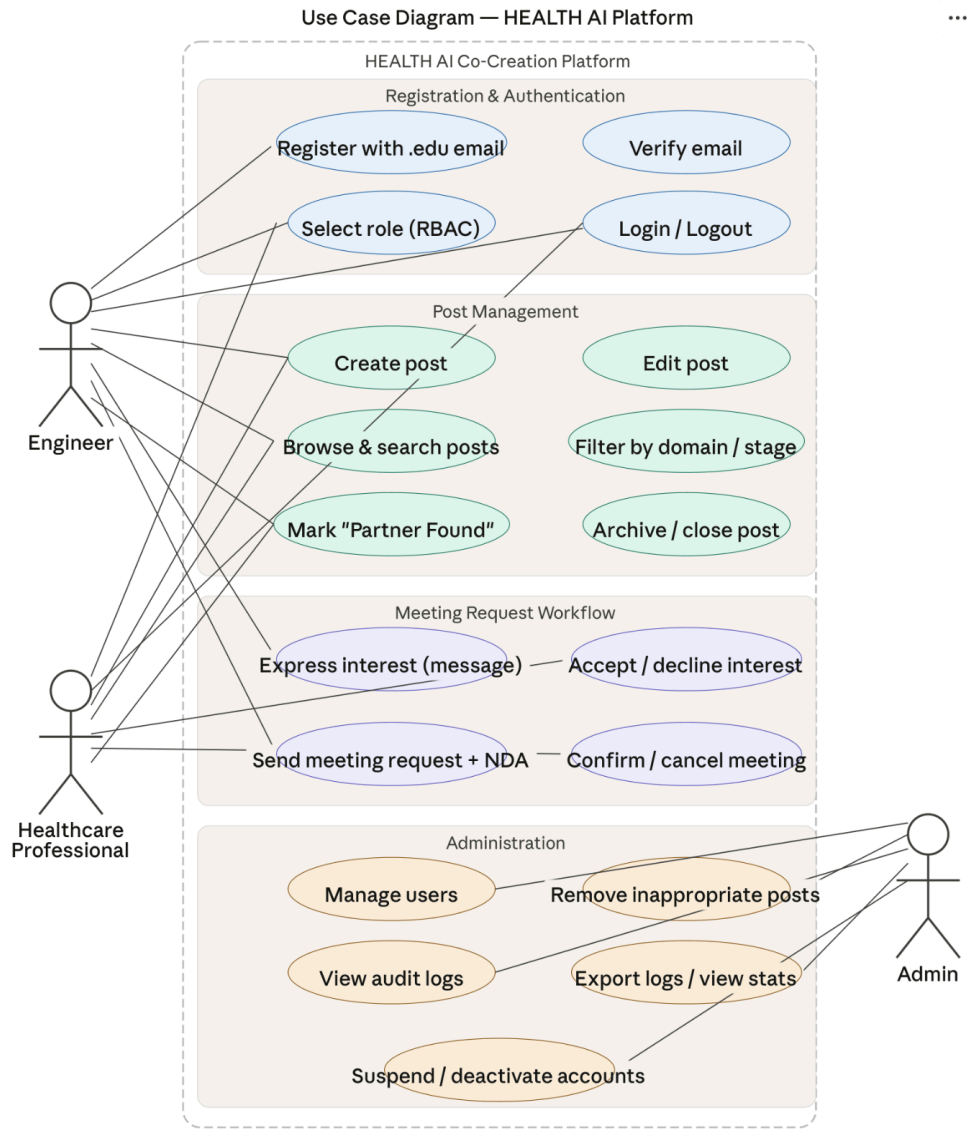
9. Appendices

The following diagrams were produced as part of the SRS preparation. See the original PDF appendix pages for the full visual artefacts.

- Use Case Diagram — covers UC-01 through UC-04 and the relationships between actors and use cases.

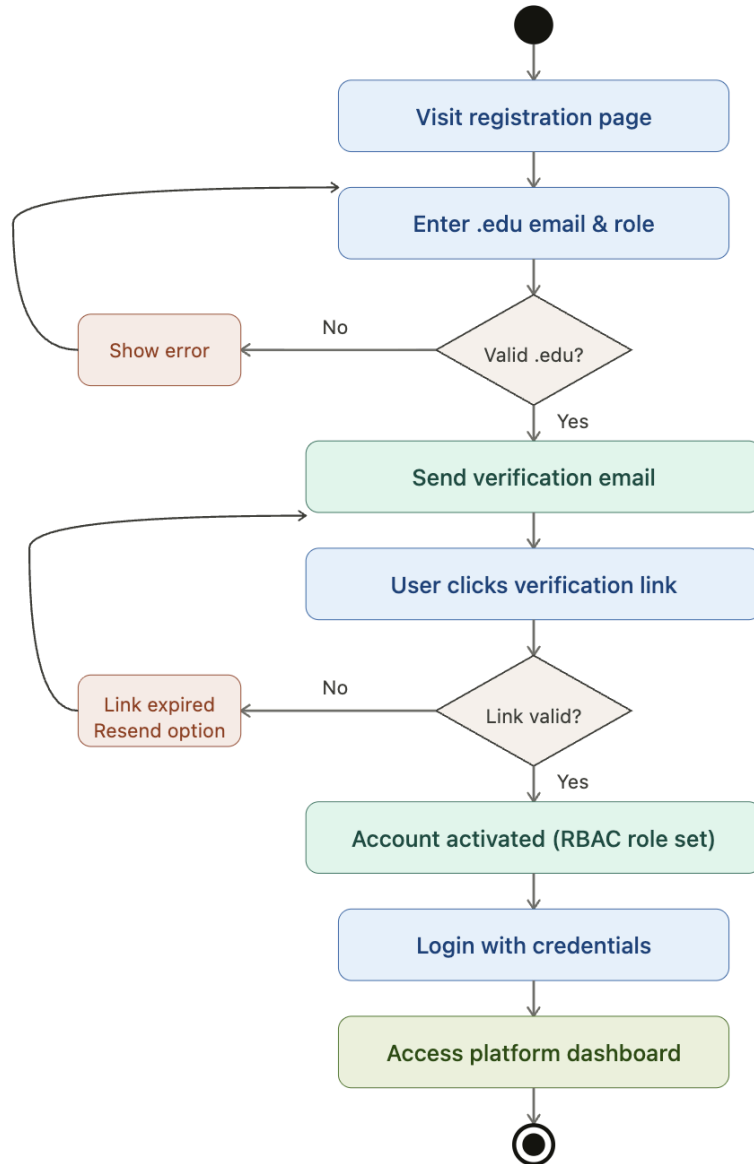
Visualize Widget

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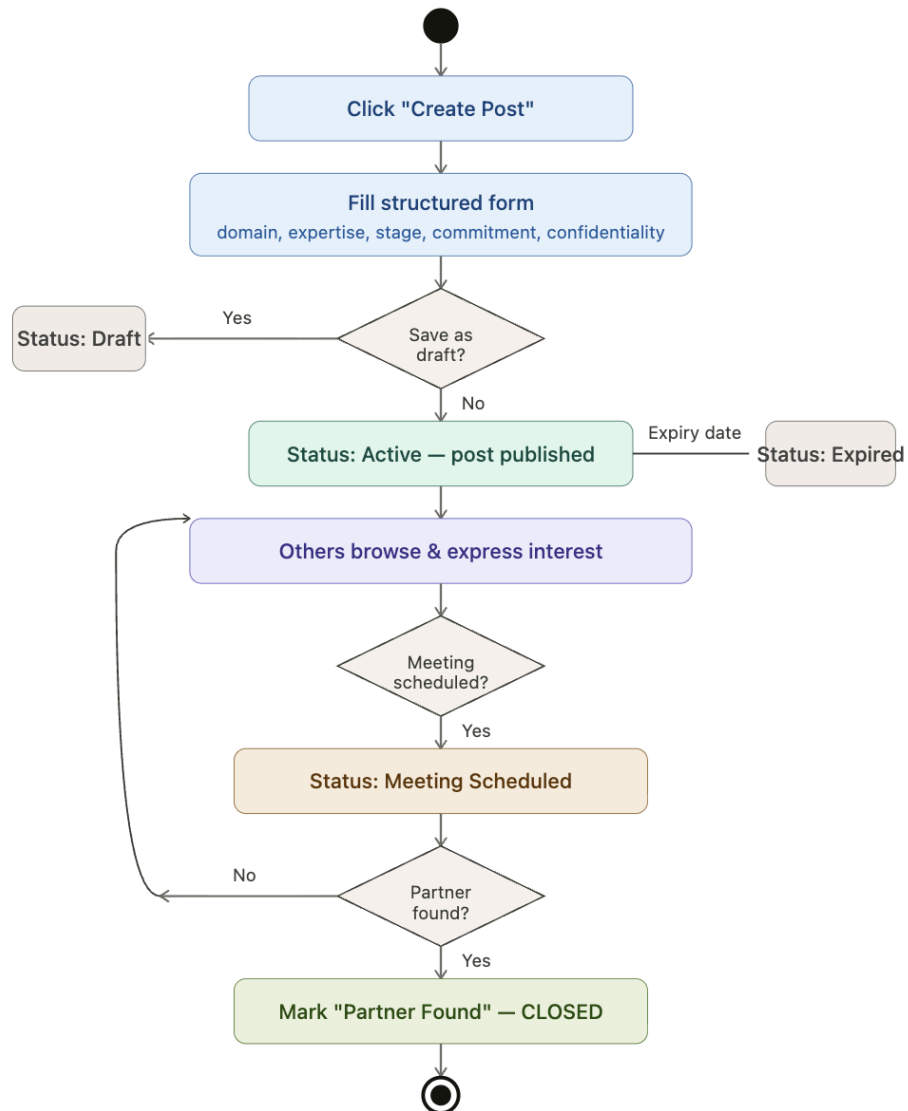


- Activity Diagrams — detail the step-by-step flows for registration/login, meeting request, and admin moderation.

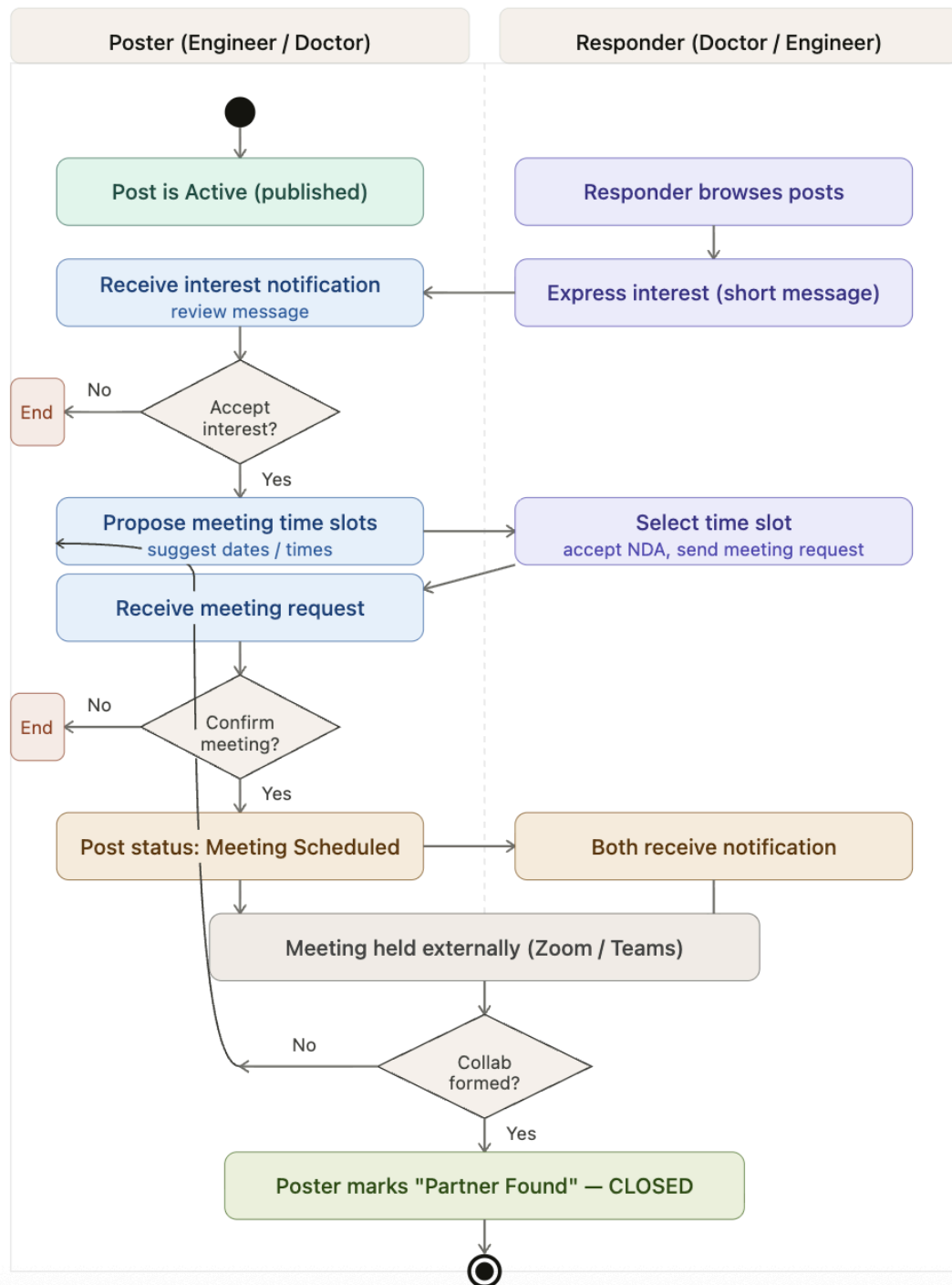
Activity Diagram 1 — User Registration & Login



Activity Diagram 2 — Post Creation & Lifecycle

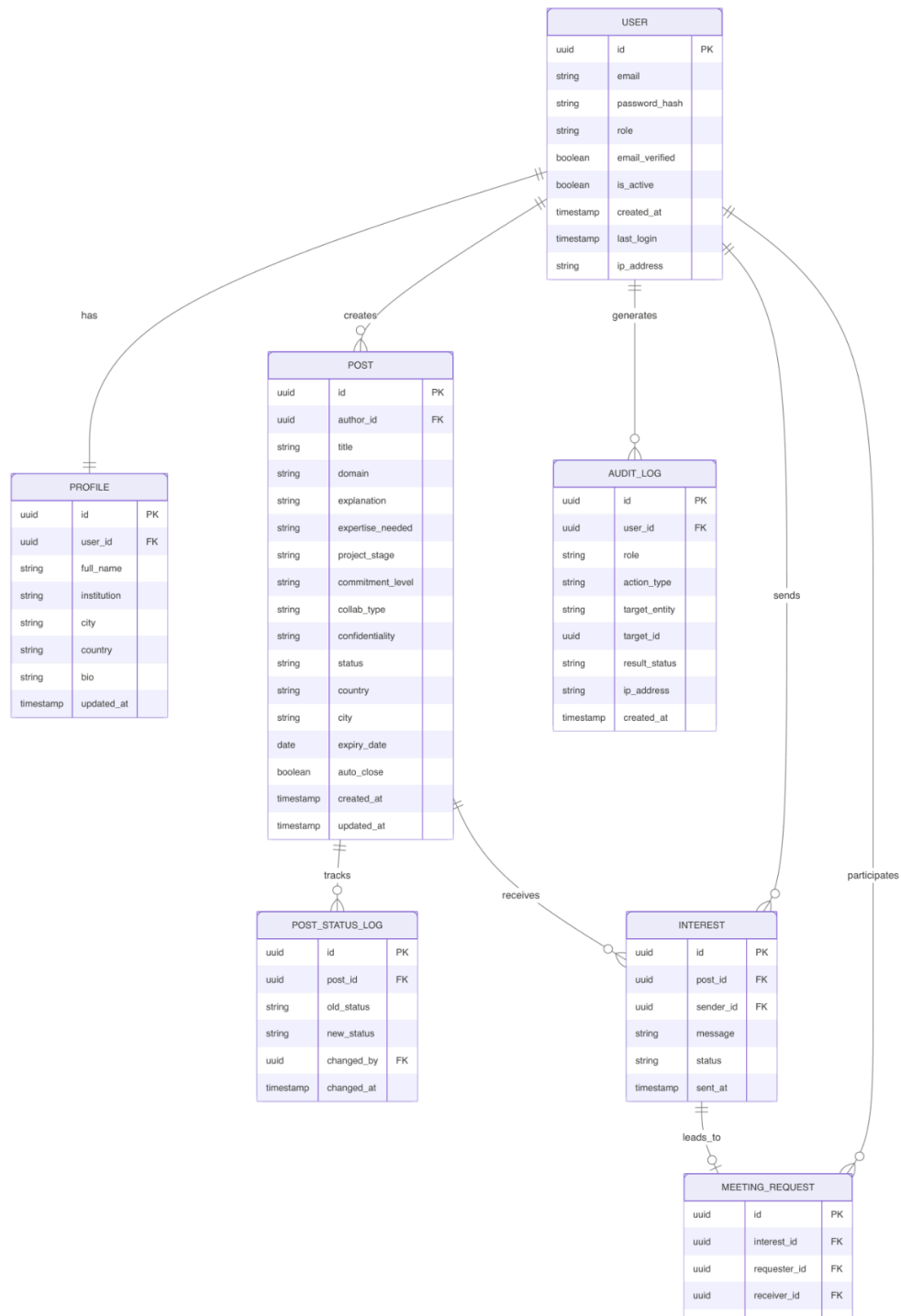


Activity Diagram 3 — Meeting Request Workflow



- ER Diagram — entity-relationship model reflecting the User, Post, MeetingRequest, and AuditLog entities described in Section 6.

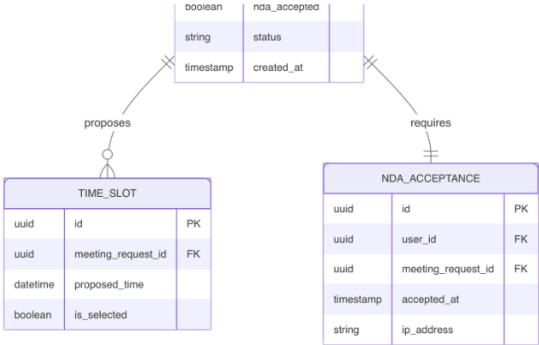
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file:///Users/cerenkizilay/Desktop/health_ai_platform_erd.html

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- Wireframes — lo-fi mockups for: Landing Page, Auth Page (Login / Register), Post Editor, Meeting Request Modal, and Admin Panel.

● HEALTH AI

Create your account

Institutional .edu email required

Full name

Dr. Ayşe Kaya

Institutional email

ayse.kaya@university.edu

Password

•••••

City

Istanbul

I am a:

Healthcare professional

Engineer

 A verification email will be sent to your address. Personal email providers are not accepted.

Create account

Already have an account? [Sign in](#)

● HEALTH AI

+ New post

AK

MENU

Dashboard

My posts

Meeting requests

Notifications

FILTER BY CITY

Istanbul

Ankara

Berlin

+ more

All

Engineer posts

Healthcare posts

Domain: any

Stage: any

AI-assisted ECG interpretation for rural clinics

Engineer

Active

Istanbul

Seeking a cardiologist or GP with experience in rural settings to validate an ML model for ECG anomaly detection. Advisor or co-founder role. Prototype stage.

Cardiology

ML / AI

Prototype

View post

Digital triage tool for oncology follow-ups

Healthcare

Active

Ankara

Oncologist looking for a software engineer with mobile app experience. Need a structured follow-up system for post-chemotherapy patients. Idea stage, high commitment.

Oncology

Mobile

Idea

View post

Smart hospital bed sensor integration

Engineer

Meeting scheduled

Berlin

IoT hardware engineer seeking ICU nurse or clinical workflow specialist. Pilot testing phase. Details shared in meeting only.

IoT

ICU

Pilot

View post

SENG384 | WebStar



Basics

2 Details

3 Settings

Post details

Describe what you're looking for. Don't include sensitive technical IP.

Title

AI-assisted ECG interpretation for rural clinics

Working domain

Cardiology imaging



Short explanation (public pitch)

Building an ML model for ECG anomaly detection targeted at rural clinics with limited specialist access. Looking for a clinical partner to validate outputs and guide ethical approval.

218 / 500

Healthcare expertise needed

Cardiologist or experienced GP, rural setting preferred

Collaboration type

Advisor

Co-founder

Research partner

Confidentiality

Public short pitch

Details in meeting only

⬅️ Back

Next: settings ➡️